

Task 1: Multithreaded counter using pthread_create() and pthread_join()

Code:

```
karthik@Karthikeya: ~/ossp/v x + v
GNU nano 7.2 race.c *
#include <stdio.h>
#include <pthread.h>

#define THREADS 4
#define ITERATIONS 1000000

long counter = 0;

void *increment(void *arg) {
    for (int i = 0; i < ITERATIONS; i++) {
        counter++;
    }
    return NULL;
}

int main() {
    pthread_t threads[THREADS];

    for (int i = 0; i < THREADS; i++)
        pthread_create(&threads[i], NULL, increment, NULL);

    for (int i = 0; i < THREADS; i++)
        pthread_join(threads[i], NULL);

    printf("Expected counter: %d\n", THREADS * ITERATIONS);
    printf("Actual counter: %d\n", counter);

    return 0;
}
```

Output:

```
karthik@Karthikeya: ~/ossp/v x + v
karthik@Karthikeya:~$ mkdir -p ~/ossp/week11
karthik@Karthikeya:~$ cd ~/ossp/week11
karthik@Karthikeya:~/ossp/week11$ nano race.c
karthik@Karthikeya:~/ossp/week11$ gcc race.c -o race -pthread
karthik@Karthikeya:~/ossp/week11$ ./race
Expected counter: 4000000
Actual counter: 1127465
karthik@Karthikeya:~/ossp/week11$ |
```

Task 2: Fix the race condition using a mutex

Code:

```
karthik@Karthikeya: ~/ossp/v × + v
GNU nano 7.2 mutex_counter.c *
#include <stdio.h>
#include <pthread.h>

#define THREADS 4
#define ITERATIONS 1000000

long counter = 0;
pthread_mutex_t lock = PTHREAD_MUTEX_INITIALIZER;

void *increment(void *arg) {
    for (int i = 0; i < ITERATIONS; i++) {
        pthread_mutex_lock(&lock);
        counter++;
        pthread_mutex_unlock(&lock);
    }
    return NULL;
}

int main() {
    pthread_t threads[THREADS];

    for (int i = 0; i < THREADS; i++)
        pthread_create(&threads[i], NULL, increment, NULL);

    for (int i = 0; i < THREADS; i++)
        pthread_join(threads[i], NULL);

    printf("Expected counter: %d\n", THREADS * ITERATIONS);
    printf("Actual counter: %ld\n", counter);

    pthread_mutex_destroy(&lock);
    return 0;
}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File ^_ Replace   ^U Paste     ^J Justify  ^_/ Go To Line M-E Redo     M-6 Copy
```

Output:

```
karthik@Karthikeya:~/ossp/week11$ nano mutex_counter.c
karthik@Karthikeya:~/ossp/week11$ gcc mutex_counter.c -o mutex_counter -pthread
karthik@Karthikeya:~/ossp/week11$ ./mutex_counter
Expected counter: 4000000
Actual counter: 4000000
karthik@Karthikeya:~/ossp/week11$ |
```